Proof of Mind

A Value Chain for Verified Mental Contribution

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Abstract

Bitcoin is widely imagined as a value chain: a ledger on which doing valuable work earns provable, accruing, transferable value. It is not. Bitcoin is a *possession* chain. It records who holds which token, orders blocks by burned energy, and lets an external market price the token; the "work" in proof-of-work is decoupled from any useful output. This paper specifies the system the folk myth describes: a chain on which value is *created* as units of contribution, *measured endogenously* by a cooperative-game value function over real outcomes, *owned and transferred* in the manner of unspent transaction outputs, and used to *secure consensus*, by *Proof of Mind* (PoM): a proof of verified, synergy-weighted mental contribution in place of a proof of wasted energy. We give the architecture (provenance-complete blocks, Bitcoin-shaped ownership, a synergy value aggregated by the Myerson value over a provenance graph, elicited by a learned reward model and estimated by permutation sampling), the consensus rule (PoM-weighted, with a participation-aware finalization basis and a coalitional stability constraint), and the mapping from Theory of Mind to Proof of Mind that makes a decentralized network of minds tractable. We are explicit about the one hard problem a value chain cannot sidestep, trustworthy and un-gameable value *measurement*, and we treat it as the load-bearing open problem rather than a solved one. The mechanisms reported here are implemented and tested in an open reference node; we mark throughout what is demonstrated and what is designed.

1 Introduction: the value chain Bitcoin is mistaken for

Proof-of-work secures *order*, not *value*. A miner proves it expended energy; the network agrees on a transaction ordering; the coin's worth is set off-chain by a market. Nowhere does the chain record *who contributed what value*. The popular intuition, that miners do work and earn value in proportion to it, is a category error: the work is arbitrary, and the value is exogenous.

A genuine value chain would instead record contribution itself. Each unit of value created, measured by what it actually added, attributed to its creator, transferable, and load-bearing for consensus. The reason such a chain is rare is not ideology but difficulty. A possession chain never has to *measure* value, because the market does; a value chain must. Measuring contribution objectively and un-gameably is the hard part, and it is the part this paper takes seriously.

Proof of Mind is our answer. Its lineage is a single line of descent: proof-of-work, then proof-of-*useful*-work (information density as work), then **proof of mind**, a proof of verified, synergy-weighted contribution, as the apex. The unit of work is a block of thought; the proof is that the thought contributed measurable value. Figure 1 shows how one unit of contribution becomes both consensus weight and tradable state.

*Written with JARVIS, an AI research collaborator. The collaboration is itself an instance of the contribution economy this paper specifies: a human and a machine producing provenance-complete, attributable work.

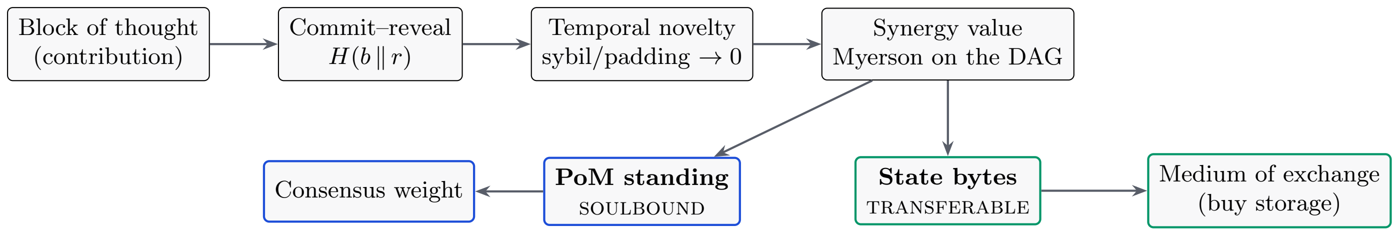

Figure 1: The contribution pipeline. A signed, provenance-complete block is ordered by commit–reveal, priced by temporal novelty, aggregated by a synergy value function, and split into a soulbound standing (consensus weight, cannot be sold) and a transferable state-capacity (a tradable commodity). You can buy storage; you cannot buy consensus.

2 Blocks and provenance

A block is the unit a participant produces, $b = \{\,\mathrm{id},\ \mathrm{parent},\ t,\ \mathrm{inputs},\ \mathrm{output},\ h\,\}$. Inputs are retained so that *how* the output came about is provable and reproducible: provenance, not merely possession. Authorship is made un-front-runnable by **commit–reveal**. A producer publishes a commitment

$$c \;=\; H(b \,\|\, r) \tag{1}$$

with a signature and timestamp *before* revealing the content b and the secret r. This binds authorship and ordering before disclosure, the same mechanism a fair batch exchange uses to eliminate front-running. Committing and then failing to reveal valid content is slashable. Crucially, the commit order is not only an anti-front-running device: it supplies the canonical ordering that makes the value rule of Section 4 strategyproof.

3 Ownership: Bitcoin-shaped, recursive for co-authorship

Each block is locked to an owner public key. Only the current owner's key produces a valid attestation, and ownership is transferable: the owner signs a reassignment to a new key, exactly as a UTXO is spent to a new locking script. Current ownership is not stored as a mutable table; it is *derived* by folding a signed transfer log over a genesis owner,

$$\mathrm{owner}_t \;=\; \mathrm{fold}\big(\mathrm{owner}_0,\ [\,\tau_1, \tau_2, \ldots, \tau_t\,]\big), \tag{2}$$

so there is no ownership record to forge; there is only a signature chain to verify. Transfer voids the prior attestation, and the new owner must re-sign.

A real block often has several authors: a prompt, a generation, a correction. Ownership is then a *share vector*, a set of (contributor, share) pairs, like a multi-output transaction. The within-block shares are computed by the *same value mechanism one level down*: the contributors are players in an intra-block coalition game, and their shares are the Myerson value of their marginal contributions to that block's output. The economy is two-level recursive, outcome $\to$ blocks $\to$ contributors, and PoM sums a contributor's shares across all blocks.

4 Endogenous value: an honest, synergy-aware price

The central object is a value function $v(S)$ over sets of blocks. Everything else, the score, the stake, the consensus weight, is a functional of v. The difficulty is that the obvious choices of v are either trivial or gameable, and the contribution of this section is a v that is neither.

4.1 Elicitation: from pairwise judgment to cardinal value

Human and model judgments arrive as *pairwise* preferences: block i contributed more than block j. Under the Bradley–Terry model each item has a latent strength s_i and

$$\Pr[\, i \succ j \,] \;=\; \frac{e^{s_i}}{e^{s_i} + e^{s_j}} \;=\; \sigma(s_i - s_j), \tag{3}$$

which is logistic regression on the score difference; the maximum-likelihood fit is convex and unique whenever the comparison graph is strongly connected. Generalized from a fixed set of items to a learned function $r_\theta(x)$ of block features, the same objective

$$\mathcal{L}(\theta) \;=\; -\sum_{(i,j)\,:\, i \succ j} \log \sigma\big(r_\theta(x_i) - r_\theta(x_j)\big) \tag{4}$$

is exactly the loss used to train a reinforcement-learning-from-human-feedback reward model. The distilled judgment model and the reward model are the same object, and that object is the production evaluator for v.

4.2 The additivity trap

A natural first attempt sets a block's value to its share of pairwise wins. We built this, and it fails in an instructive way. If the coalition value is additive, $v(S) = \sum_{i \in S} v(\{i\})$, then the Shapley value collapses to the standalone value, $\phi_i = v(\{i\})$, and the normalized allocation is just the proportional (Copeland) win-share.

Claim 1. *For an additive game the entire $2^{|N|}$-coalition Shapley computation returns the same numbers as a one-line normalization. The cooperative machinery does no work.*

Pairwise comparison yields a *ranking*, which is additive by construction, so Shapley over it is ceremonial. Value measurement is meaningful only when $v(S)$ has **synergy**: a coalition's worth differs from the sum of its members' standalone worth, through complementarity, redundancy, or pivotality.

4.3 Synergy and the Myerson value

We therefore define $v(S)$ as an *outcome value*: the quality or completeness of the outcome reconstructable using only the blocks in S. A block the outcome fails without is pivotal and earns a high marginal value; a block redundant with others earns little. Credit is the **Myerson value**, the Shapley value of the graph-restricted game in which only coalitions connected in the provenance graph G create value:

$$\phi_i(v) \;=\; \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[\, v^G(S \cup \{i\}) - v^G(S) \,\big], \qquad v^G(S) = \sum_{T \in S/G} v(T), \tag{5}$$

where S/G are the connected components of S in G. Value flows along provenance edges, not over arbitrary sets; plain Shapley is the special case of a complete graph. Exact computation is exponential, so we estimate ϕ_i by Monte-Carlo permutation sampling (Data-Shapley), which is Shapley specialized to data valuation. In a working prototype over real blocks, replacing the additive game with a submodular coverage outcome-value made the Myerson values diverge materially from the proportional baseline, with mean absolute difference $\|\phi - \text{Copeland}\|_1 \approx 0.26$: the cooperative value became load-bearing, rewarding pivotal blocks and discounting redundant ones.

4.4 Strategyproofness: temporal novelty

A value function over content is gameable by repetition unless novelty is priced. We use the commit order to do exactly that. The value of a block is its coverage *minus* the coverage already claimed by all earlier-committed blocks,

$$\nu(b) \;=\; \Bigl|\; \mathrm{cov}(b) \;\setminus \bigcup_{c\,:\,t_c<t_b} \mathrm{cov}(c) \;\Bigr|. \tag{6}$$

First-to-reveal earns the novelty; later padding or sybil copies earn zero. This is why commit–reveal is load-bearing for *value*, not only authorship: it provides the canonical order that makes the rule strategyproof. In the reference implementation a sybil ring of five identical-content identities banks the coverage of at most one block (1.06×, not 5×), and a cross-block re-post earns zero.

4.5 Saturation under coordinated volume

Novelty defeats duplication, but a subtler attack remains: a vested participant, or several colluding ones, can post many distinct-but-low-value children of a valuable parent and try to pump the parent's aggregated flow. We damp this with a single geometric decay over the global, canonically-ordered list of a parent's contributing children. With decay $\rho = 1/\varphi$ (the inverse golden ratio), the parent's aggregated contribution is

$$F \;=\; \sum_{j\geq 0} \rho^j \, c_{(j)} \;\leq\; \frac{c_{\max}}{1-\rho} \;=\; \varphi^2 \, c_{\max} \;\approx\; 2.618\, c_{\max}, \tag{7}$$

a convergent series: coordinated volume *saturates* rather than amplifies. An earlier design that damped two axes separately (within an identity and across identities) left their product unbounded, a "diagonal" pump in which K coordinated identities each posting M children compounded both tails. Folding both into one joint decay over the flattened global order closes it. Figure 2 shows the measured effect: before the fix the diagonal grows past the honest single-identity ceiling; after it, the whole grid is bounded by that ceiling.

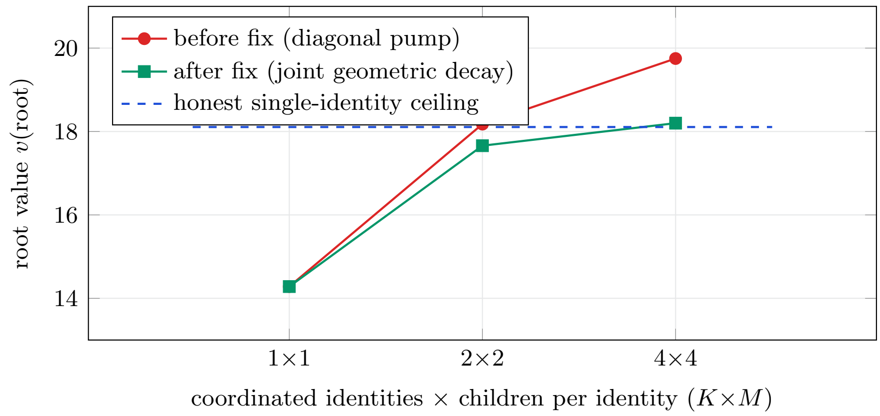

Figure 2: Coordinated-volume saturation, measured in the reference node. Before the joint-decay fix, K coordinated identities each posting M valueless children pump the root's aggregated value past the honest single-identity ceiling (red). After folding the two damping axes into one geometric decay over the global canonical order (green), the entire K×M grid is bounded by that ceiling to within 2%. Honest single contributions are paid unchanged.

4.6 Certifying the value: the HodgeRank residual

Pairwise judgments need not be transitive; intransitivity is inherent (Condorcet), not merely noise. A scalar value function is the right object only when the judgments are close to a gradient flow. We make this checkable. Encoding comparisons as a skew-symmetric edge flow Y on the comparison graph and solving the weighted least-squares problem

$$\min_{s} \sum_{i,j} w_{ij} \left(s_i - s_j - Y_{ij}\right)^2 \tag{8}$$

yields a scalar ranking s (a graph-Laplacian solve, generalizing Borda). The Helmholtz–Hodge decomposition splits the flow orthogonally,

$$Y = \underbrace{\operatorname{grad} s}_{\text{global ranking}} \oplus \underbrace{\ker \Delta_1}_{\text{harmonic (global cycles)}} \oplus \underbrace{\operatorname{curl}^*}_{\text{local cycles}} . \tag{9}$$

The residual energy is not an error bar; it is a *certificate*. A large curl component says fine-scale ordering is unreliable; a large harmonic component says the judgments cycle through holes in the graph. Where the Shapley/Myerson construction *assumes* a consistent scalar world (its axioms force the residual to zero), HodgeRank *measures* whether that world holds, and localizes where it breaks. This matters for security as well as honesty: a coordinated sybil or collusive ring injects detectable harmonic circulation, so a spike in harmonic energy is a manipulation alarm in the same computation that produces the value. The value layer therefore ships its own certificate of when the single number it reports can be trusted.

4.7 The learned evaluator, bounded by construction

The production $v(S)$ is a learned reward model, and learned models drift and can be probed. We do not ask it to be un-gameable. We deny it the authority to be dangerous. The strategyproof skeleton is the realized gate

$$\text{seed}_i = \bar{\nu}_i \cdot g(f_i) \cdot \omega(\ell_i), \qquad \bar{\nu}_i,\ g(f_i),\ \omega(\ell_i) \in [0,1], \tag{10}$$

the product of floored novelty $\bar{\nu}_i$, a realized-flow gate $g(f_i)$, and the learned outcome ω scored on the block's provenance lineage ℓ_i. Because the factors are multiplicative and bounded in $[0,1]$, each can only *lower* the seed; none can mint. The learned model is confined to two roles that have no minting power: it may offer an honest contributor an early liquidity *advance* against expected vesting (its own capital at risk, repaid from realized value and slashed on shortfall), and it may enter a dispute as *evidence* the jurors weigh. A corrupt model that scores everything maximally reduces to the un-learned gate exactly: in the reference node, $v_8 \equiv v_7$ under an adversarial $+\infty$ model. The proof obligation collapses from "the neural network is un-gameable" to "three bounds hold," which are small functions with tests.

The value of the learned model is what it adds *within* those bounds. Figure 3 reports a held-out measurement: trained on one split of coalitions and tested on coalitions never seen, the learned $v(S)$ ranks "work built on work" above the same cells dumped as orphans with accuracy ≥ 0.9, while the coverage proxy is blind to lineage and ties at chance.

5 Measurement as a living mechanism

The value function of Section 4 is not a fixed formula, and it cannot be one. Any static value rule is Goodhart-bait: the moment it is published, it is gamed, because a fixed objective tells an adversary exactly what to optimize. The only un-gameable measure is one that adapts to the gaming. Measurement on this chain is therefore a closed loop, not a closed form, and its components, introduced separately above, are one adaptive system.

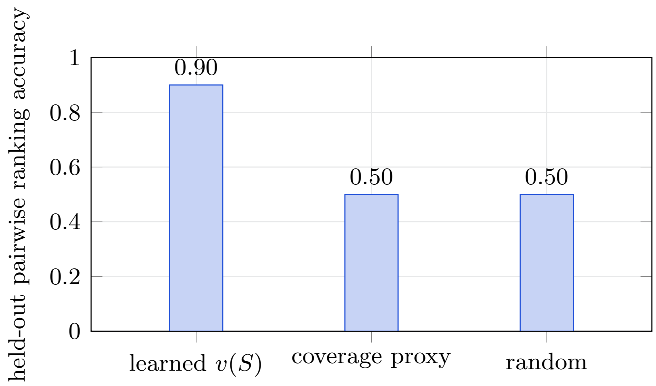

Figure 3: The moat, measured not asserted. On coalitions never seen in training, the learned $v(S)$ distinguishes connected, synergistic contribution from the same content as disconnected orphans (≥ 0.9), while the coverage proxy the per-block rule can see is blind to lineage and ranks at chance (0.5, red line). The remaining mile is real outcome-preference data; the measurement harness runs unchanged when it lands.

1. **Realized, not predicted.** The learned model can only lower a seed it never raises one (Section 4). Value is gated on realized downstream flow, so the measure is anchored to what actually happened, not to a prediction an adversary can spoof.
2. **It retrains on outcomes.** The chain is a clean, provenance-complete, value-weighted dataset (Section 9). The evaluator is fine-tuned on verified high-value contributions against caught failures, so the measure sharpens as the chain accumulates verified truth. Yesterday's attack becomes today's training signal.
3. **Value is provisional, then adjudicated.** A payout is not final at emission. A challenge window lets a bonded challenger refute a value claim; a refuted claim is zeroed and the claimant slashed. Measurement is a dispute process with finality, not a one-shot score.
4. **It cannot be banked.** Standing decays, so a high score earned once does not persist without continued contribution. The measure tracks live value, and a stale position is reclaimed.
5. **It diagnoses its own breakdown.** The HodgeRank residual (Section 4) is a continuous detector: when the judgments stop fitting a single scalar, through genuine multidimensionality or through a coordinated manipulation injecting harmonic circulation, the residual energy rises and flags it. The system therefore knows *when* to distrust its own number, rather than reporting a confident scalar over a structure that no longer supports one.
6. **The dimensions of value can themselves evolve.** The value matrix is governance-mutable but verifier-gated: a new dimension is admitted only if it predicts realized downstream value, weights move within bounded ranges, and no real dimension can be zeroed. The measure is not frozen at genesis, and it cannot be captured into a private one.

Put together, the loop is: measure, realize, dispute, retrain, decay, re-measure. A static rule loses to Goodhart on its first day; an adaptive measure with an un-gameable realized-value anchor, a dispute process for finality, and a residual that reports its own validity does not present a fixed target to optimize against. This is why Section 10 calls value measurement the load-bearing open problem rather than a solved one. The claim is not a closed-form proof of un-gameability; it is a mechanism that treats un-gameability as a moving equilibrium it must continuously hold, and whose present strength is measured (Figure 3) rather than asserted.

6 Proof of Mind

A participant's **PoM score** is its accumulated Myerson value over the verified, owned, provenance-complete blocks it holds,

$$\mathrm{PoM}(a) \;=\; \sum_{b\,\in\,\mathcal{B}(a)} \phi_b, \qquad \mathcal{B}(a) = \{\text{verified, owned, provenance-complete blocks of } a\}. \tag{11}$$

Three properties are inherited from the construction. It is **verifiable**: every contributing block is signed and provenance-complete, so anyone can recompute the score from public data. It is **sybil-resistant structurally**: PoM requires owned blocks whose value was synergy-judged, and splitting one mind across many accounts does not multiply PoM because the synergy game discounts redundant copies; a thousand empty nodes score zero. And it is **earned, not bought**: the stake is accumulated proof of contribution, and slashing revokes PoM for refuted blocks, caught hallucinations, and failed attestations.

6.1 Theory of Mind to Proof of Mind

Across agents, Theory of Mind is inference. You cannot see inside another mind, so you guess whether to trust it. That guess is an airgap, the inter-mind version of the gap between a blockchain and reality. Economic Theory of Mind reframes a mind as an economy whose outputs carry value. Proof of Mind is the verifiable economic proof of that economy, and it closes the airgap: instead of each node inferring whether another is a mind worth trusting (intractable and game-able), PoM supplies a proof anyone can recompute. Trust stops being a per-mind inference and becomes a structural property. The chain is capacity → mind-as-economy → proof of that economy.

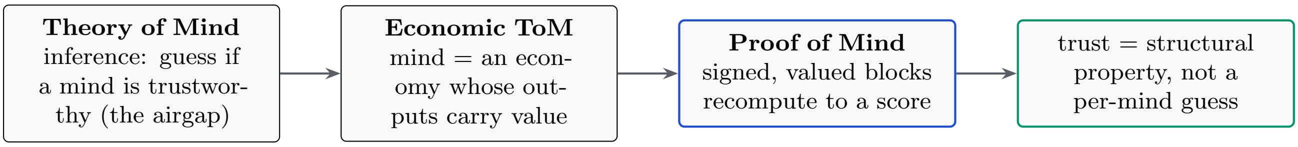

Figure 4: The mapping that makes a decentralized network of minds tractable: replace "infer whether to trust" with "verify a proof."

7 Consensus

Validators are weighted by **PoM**, not by energy or capital. The tamper-evident, signed, owned chain is the ledger; PoM is the stake; consensus is PoM-weighted agreement on the canonical chain. Three design choices make this safe.

7.1 Retention decay and the finalization basis

Vote weight decays with staleness, so standing tracks live participation rather than past glory. Decay applies to the *franchise*, the vote weight, never to the staked balance, and symmetrically across all dimensions so the effective mix is time-invariant. The remaining question is what a supermajority is a fraction *of*. A base-weight denominator is eclipse-resistant but halts under low participation; a pure effective-weight denominator removes the halt but lets an attacker shrink the denominator and finalize a minority. The stable choice is the hybrid

$$\text{finalize} \iff \forall d: \quad W^d_{\text{for}} \;\geq\; \tfrac{2}{3}\,\max\big(W^d_{\text{eff}},\, Q^d\big), \tag{12}$$

which pins the denominator at a quorum floor Q^d, a fraction of base weight, so neither failure occurs. The composition over the dimensions d is AND-gated, not a substitutable weighted sum:

a single dimension below the supermajority cannot finalize alone. This is verified against tested reference models, including a self-audit that demonstrates the eclipse on a bare effective basis and its closure under the quorum floor.

7.2 Stability: no profitable fork

PoM-weighted agreement must be defection-proof; no coalition of validators should profit by deviating. We impose a coalitional stability constraint over the PoM-weighted game: require the allocation to lie in the **core** when it is non-empty (no sub-coalition does better by forking), and fall back to the **nucleolus** when the core is empty (the unique allocation that lexicographically minimizes the worst coalition's incentive to defect). This is added precisely because consensus requires it; pure attribution does not need a stability concept.

7.3 Three powers

Consensus is not a single weighted vote. It is the agreement of three independent powers, **cognition** (PoM), **capital** (state-stake), and **compute** (proof-of-work, carried by a separate money layer), composed by conjunction. Three is the minimum for a non-dominated cyclic equilibrium: with two powers one is the strict best response and captures the other; with three the relation is non-transitive (rock–paper–scissors) and capture-resistant; four or more adds complexity without more non-domination. Because the composition is an AND of independent layers rather than a weighted sum, a large share in one dimension is a reward weight, not a vote that can finalize alone. An attacker must defeat all three.

8 Cryptoeconomics: one PoM, one byte

Storage is the scarce resource, and PoM is the right to occupy it: **one PoM equals one byte of on-chain state**. Issuance is earned, not bought, because novel verified contribution is the mint; state-rent is PoM decay, which both reclaims idle capacity and serves as the supply sink that bounds total live PoM. The mint (novel contribution) and the burn (decay) balance.

The subtlety that keeps this from collapsing into proof-of-stake is a split between two cells. A participant's **standing** is soulbound: it is the franchise (consensus weight and the right to mint state), and it cannot be sold. The **state-capacity** it earns the right to mint is transferable: a byte of state is a liquid commodity that trades freely. The result is that you can buy storage but not consensus. A transferable franchise would let a rich actor buy consensus, which is proof-of-stake by another name; a soulbound franchise makes "earned, not bought" and structural sybil-resistance real at once.

9 Backwards-enforcement of the model

The value chain is also a training signal. Each block is provenance-complete, owner-authenticated, and Myerson-valued: a clean, value-weighted dataset, exactly the object Data-Shapley attributes. With open model weights, fine-tuning on high-PoM verified blocks against caught hallucinations lets the governance layer's accumulated, verified truth shape the model. With closed weights, the same truth constrains the model in-context: gates block disallowed actions and the signed chain contradicts a hallucination, forcing correction. Governance produces a training signal, which shapes the model, which governance then verifies, and the loop compounds. The structure does not only constrain the mind; it teaches it.

10 Security and honest limitations

The load-bearing bet. A value chain must measure value un-gameably, the problem Bitcoin sidesteps by being mere possession. If the learned outcome evaluator is gameable, PoM degrades into a reputation system. This is the central open problem, and we treat it as such. The mitigations are structural: the realized-flow gate that the learned model can only lower, the temporal-novelty rule, the HodgeRank manipulation certificate, and the held-out measurement of Figure 3. The remaining mile is real outcome-preference data at scale.

Self-reference. A block can carry information about contributions and thereby influence the scoring of others. The value layer scores such a block for its own marginal contribution; the elicitation layer consumes its content as a judgment about referenced blocks. These must stay decoupled, and no block may be the sole scorer of its own value. Guards: independent evaluators, eigenvector-style damping over the attribution graph, and the synergy game's discounting of self-referential coalitions.

Demonstrated versus designed. *Demonstrated* in the open reference node (a Rust implementation with a no-std verification core shared with on-chain scripts, 280 passing tests at the time of writing): ownership and transfer, per-block signing, tamper-resistance, the synergy value over a coverage proxy with Myerson and Bradley–Terry and permutation sampling, temporal novelty, coordinated-volume saturation (Figure 2), the bounded learned evaluator and its held-out measurement (Figure 3), the dispute and slashing path, PoM-weighted finalization with the hybrid basis, the core/nucleolus stability solver, and the on-chain recomputation of the ordering and finalization rules inside the virtual machine. *Designed, not yet built*: the learned $v(S)$ on real outcome labels, the production two-level recursion, owner-signature control of spends (the deterministic transaction digest the signature will cover is built and tested), the open-weight fine-tune loop, and the money/stability layer.

11 Fair launch and theft-resistance

The creator's pre-launch contribution would otherwise be an unearned head start. It is neutralized by a **genesis-burn**: the chain stays continuous from genesis and pre-launch blocks remain auditable, but their standing and state-value are programmatically burned to zero at the launch height. This is a fair launch one can verify on-chain, chosen over a chain reset, which only asserts fairness.

The mechanism also yields an unusual theft-resistance. Because the system *is* attribution, a derivative deployment has two options. It can strip the provenance, in which case the attribution layer the system depends on is gone and the fork is incoherent; or it can preserve the provenance, in which case its edges trace back to this origin and running it credits the source. Copying the network honestly adds the copier to the same attribution graph, so there is no outside to fork to. Forking, which fragments value on a possession chain, here becomes contribution.

12 Conclusion

We have specified a chain on which value is created as contribution, measured endogenously by a synergy-aware cooperative value over a provenance graph, owned and transferred like a UTXO, and used to secure consensus through Proof of Mind. The design replaces the inferences that make decentralized trust intractable, "is this a mind worth trusting," with proofs anyone can recompute, and it replaces proof of wasted energy with proof of verified thought. The honest center of the system is a single hard problem, un-gameable value measurement, and the architecture is built so that the learned component charged with it can only ever deny value, never mint it, while a separate certificate reports when the value can be trusted at all. What remains is to feed that evaluator real outcomes at scale. The rest is assembled from known mathematics, and it runs.

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